

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q17-0910

East Watkinsville 100 kV  
January 16, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0910. The Interconnection Customer proposes a 52.8 MW solar photovoltaic facility interconnecting at the East Watkinsville 100 kV substation in the Illinois area. The request was received on December 29, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 383 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,210,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                   | Contingency                        | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------------------------------|------------------------------------|-------------|--------------|----------|
| TROY ENERGY LLC-DIXON LANDING SUBSTATION 230 kV Line | TROY ENERGY LLC-RICHLAND 500 kV    | 209.2       | 558          | 8.6      |
| MADELINE-BARAGA VLG 230 kV L                         | BARAGA VLG-GRAY'S FLAT 230 kV      | 145.1       | 514          | 6.5      |
| CORRIDOR-LARSON ST 230 kV                            | TREGO-BEAVER 500 kV                | 134.1       | 640          | 9.9      |
| ADIRONDACK-JAMES DE YOUNG 230 kV                     | ST. LAWRENCE-JAMES DE YOUNG 230 kV | 131.5       | 748          | 7.4      |
| EAST METHUEN 500/230kV Transformer                   | TREGO-BEAVER SUBSTATION 500 kV     | 131.2       | 1,605        | 16.0     |
| GRAY'S FLAT SUBSTATION-E METHUEN 230 kV              | E METHUEN-MADELINE 230 kV Line     | 130.5       | 808          | 8.9      |
| WALDO 500 kV/230 kV TX                               | TREGO-BEAVER 500 kV line           | 125.8       | 881          | 9.8      |
| HEBO-BURRIS PIT 230 kV                               | BEAVER-TREGO 500 kV                | 125.1       | 933          | 7.9      |
| JAMES DE YOUNG-BURRIS PIT 230 kV                     | TREGO-BEAVER 500 kV                | 124.4       | 955          | 7.9      |
| TREGO 230/1kV XFMR                                   | TREGO-TROY ENERGY LLC 500 kV       | 122.3       | 644          | 12.9     |
| CARLEY RD-JAMES DE YOUNG 230 kV                      | E METHUEN 500/230 kV Auto          | 122.3       | 647          | 7.1      |
| WINTHROP-HEBO 230 kV line                            | HEBO-BARRETT 230 kV                | 122.0       | 629          | 5.5      |
| BAAGA VLG-GRAY'S FLAT 230 kV line                    | MADELINE-EAST METHUEN 230 kV       | 121.4       | 1,020        | 8.9      |
| ADIRONDACK 500-230 kV Autotransformer                | E METHUEN 500-230 kV XFMR          | 121.3       | 1,026        | 5.1      |
| MADELINE-EAST METHUEN 230 kV                         | BARAGA VLG-GRAY'S FLAT 230 kV      | 119.7       | 1,074        | 9.5      |

| Monitored Facility                               | Contingency                                      | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------|--------------------------------------------------|-------------|--------------|----------|
| JOSO-ONANCOCK DIESEL GENERATION FACILITY 230 kV  | BEAVER SUBSTATION-EUFAULA 500 kV line            | 116.9       | 1,721        | 14.3     |
| CORRIDOR-LOOKINGGLASS 230 kV                     | TREGO-BEAVER SUBSTATION 500 kV                   | 116.3       | 763          | 9.9      |
| TREGO-HIGHGATE 230 kV L                          | TROY ENERGY LLC-TREGO 500 kV L                   | 116.3       | 511          | 10.9     |
| BEAVER-TREGO 500 kV                              | ADAMS-ONANCOCK DIESEL GENERATION FACILITY 230 kV | 116.1       | 4,242        | 30.9     |
| ADAMS-ONANCOCK DIESEL GENERATION FACILITY 230 kV | TREGO-BEAVER 500 kV L                            | 115.7       | 2,100        | 15.9     |
| TRUMAN-LOOKINGGLASS 230 kV                       | BEAVER SUBSTATION-TREGO 500 kV L                 | 115.5       | 795          | 9.9      |
| HERALD-TWIN CITIES SUBSTATION-PICKFORD 230 kV    | BEAVER-TREGO 500 kV line                         | 115.2       | 1,391        | 9.7      |
| MASON ST-NEW DEPARTURE 230 kV                    | TREGO-BEAVER 500 kV                              | 115.1       | 1,075        | 7.8      |
| ADAMS-PICKFORD 230 kV                            | TREGO-BEAVER 500 kV                              | 115.0       | 1,089        | 8.1      |
| WALDO-TRUMAN 230 kV                              | TREGO-BEAVER 500 kV                              | 114.8       | 822          | 9.9      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                             | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| TROY ENERGY LLC-DIXON LANDING SUBSTATION 230 kV Line | Rebuild line with higher-capacity conductor | \$10,798,000  | 0% (\$51,000)         |
| MADELINE-BARAGA VLG 230 kV                           | Rebuild line with higher-capacity conductor | \$13,244,000  | 0% (\$60,000)         |
| CORRIDOR-LARSON ST 230 kV                            | Rebuild line with higher-capacity conductor | \$14,103,000  | 0% (\$65,000)         |
| JAMES DE YOUNG-ADIRONDACK 230 kV L                   | Rebuild line with higher-capacity conductor | \$35,802,000  | 0% (\$163,000)        |
| EAST METHUEN 500/230kV Auto                          | Replace transformer with higher-rated unit  | \$29,100,000  | 0% (\$133,000)        |
| GRAY'S FLAT-EAST METHUEN 230 kV L                    | Rebuild line with higher-capacity conductor | \$52,108,000  | 0% (\$237,000)        |
| WALDO 500 kV/230 kV TX                               | Replace transformer with higher-rated unit  | \$13,755,000  | 0% (\$64,000)         |
| HEBO-BURRIS PIT 230 kV                               | Rebuild line with higher-capacity conductor | \$30,335,000  | 0% (\$138,000)        |
| JAMES DE YOUNG-BURRIS PIT 230 kV                     | Rebuild line with higher-capacity conductor | \$39,150,000  | 0% (\$178,000)        |
| TREGO 230/1kV XFMR                                   | Replace transformer with higher-rated unit  | \$11,003,000  | 0% (\$52,000)         |

| Facility                                         | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| CARLEY RD-JAMES DE YOUNG 230 kV                  | Rebuild line with higher-capacity conductor | \$21,821,000  | 0% (\$99,000)         |
| WINTHROP-HEBO 230 kV line                        | Rebuild line with higher-capacity conductor | \$15,427,000  | 0% (\$70,000)         |
| BAAGA VLG-GRAY'S FLAT 230 kV line                | Rebuild line with higher-capacity conductor | \$13,699,000  | 0% (\$62,000)         |
| ADIRONDACK 500-230 kV Autotransformer            | Replace transformer with higher-rated unit  | \$29,308,000  | 0% (\$131,000)        |
| MADELINE-E METHUEN SUBSTATION 230 kV             | Rebuild line with higher-capacity conductor | \$23,217,000  | 0% (\$105,000)        |
| JOSO-ONANCOCK DIESEL GENERATION FACILITY 230 kV  | Rebuild line with higher-capacity conductor | \$15,153,000  | 0% (\$70,000)         |
| CORRIDOR-LOOKINGGLASS 230 kV                     | Rebuild line with higher-capacity conductor | \$16,804,000  | 0% (\$78,000)         |
| TREGO-HIGHGATE 230 kV                            | Rebuild line with higher-capacity conductor | \$49,722,000  | 0% (\$238,000)        |
| TREGO-BEAVER SUBSTATION 500 kV line              | Rebuild line with higher-capacity conductor | \$278,939,000 | 0% (\$1,282,000)      |
| ONANCOCK DIESEL GENERATION FACILITY-ADAMS 230 kV | Rebuild line with higher-capacity conductor | \$43,247,000  | 0% (\$200,000)        |
| TRUMAN-LOOKINGGLASS 230 kV                       | Rebuild line with higher-capacity conductor | \$21,749,000  | 0% (\$101,000)        |
| HERALD-TWIN CITIES SUBSTATION-PICKFORD 230 kV    | Rebuild line with higher-capacity conductor | \$46,289,000  | 0% (\$214,000)        |
| NEW DEPARTURE-MASON ST 230 kV L                  | Rebuild line with higher-capacity conductor | \$45,498,000  | 0% (\$210,000)        |
| PCKFORD-ADAMS 230 kV                             | Rebuild line with higher-capacity conductor | \$24,348,000  | 0% (\$112,000)        |
| WALDO-TRUMAN 230 kV                              | Rebuild line with higher-capacity conductor | \$21,018,000  | 0% (\$97,000)         |
|                                                  | Total allocated to project                  |               | \$4,210,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.